

The Noncommutative Column Ball Is Not Right Regular

Candidate counterexample packet for arXiv:2411.09372
Model: GPT5.6

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Status. `candidate_counterexample_likely_valid_novelty_unconfirmed`.

1 Source question

In Example 5.9 of *Weak-* and completely isometric structure of noncommutative function algebras*, Sampat and Shalit ask whether the nc column unit ball

$$\mathfrak{C}_d = \left\{ X \in \mathbb{M}^d : \left\| \sum_{j=1}^d X_j^* X_j \right\| < 1 \right\}$$

is right regular [1]. The question appears on PDF page 23.

so $f \mapsto f \circ T$ is not an isometry, in contradiction to the first observation in the proof. ■

Example 5.8. Example 5.3 shows that $\mathfrak{D}_2$ is not right regular, and in fact does not satisfy any imaginable weaker regularity property since $\Delta f(0, X)$ need not be levelwise bounded. In fact, $\mathfrak{D}_2$ is the maximal nc operator ball with first level equal to $\mathbb{D}^2$, but since $\Delta f(0, X)$ blows up already on the first level, it follows that *no nc operator ball over $\mathbb{D}^2$ is right regular*.

Example 5.9. We do not know whether the column unit ball

$$\mathfrak{C}_d = \left\{ X \in \mathbb{M}^d : \left\| \sum_{j=1}^d X_j^* X_j \right\| < 1 \right\},$$

is right regular, but it is easy to see that it is *left* regular (in an obvious sense). This can be seen by noting that by applying left difference-differential calculus instead of right difference-differential calculus, we can write the remainder term for $f \in \mathcal{J}_N$ as

$$f(Z) = \sum_{|w|=N} g_w(Z) Z^w = \text{row}(g_w(Z))_{|w|=N} \cdot \text{col}(Z^w)_{|w|=N}.$$

Since the tuple $Z = (Z_1, \dots, Z_d)$ on $\mathfrak{C}_d$ is a column coisometry, we can argue as above and find $\|\text{row}(g_w(Z))_{|w|=N}\| = \|f\|$, etc.

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Figure 1: The source question, arXiv:2411.09372v2, Example 5.9 on PDF page 23.

For an nc operator ball $\mathbb{D}_Q$, right regularity requires, among other things, a constant M such that for

every N and every function f in the ideal generated by the monomials of length N ,

$$\left(\sup_{X \in \mathbb{D}_Q} \|\text{row}(X^w)_{|w|=N}\| \right) \left(\sup_{X \in \mathbb{D}_Q} \|\text{col}(\Delta^w f(0, \dots, 0, X))_{|w|=N}\| \right) \leq M \|f\|. \quad (1)$$

The crucial point is that M is independent of both N and f .

2 Proof intuition

The geometric factor in (1) is already too large. Although each coordinate of a column contraction is contractive, the row containing all d^N words of length N can have norm $d^{N/2}$. Finite truncated backward shifts attain this value in the limit: every word sends a distinct length- N basis vector to the same vacuum vector.

The monomial $f_N(Z) = Z_1^N$ prevents the second factor from compensating for this growth. It has norm one, and its right Taylor–Taylor coefficient column at order N contains a single nonzero entry, the constant function one. Thus the left side of (1) is exactly $d^{N/2}$.

3 Exact word-row growth

For $N \geq 1$, put

$$A_{N,d} = \sup_{X \in \mathfrak{C}_d} \|\text{row}(X^w)_{|w|=N}\|.$$

Lemma 1. *For every $d, N \geq 1$, one has $A_{N,d} = d^{N/2}$.*

Proof. If $X \in \mathfrak{C}_d$, then $X_j^* X_j \leq \sum_k X_k^* X_k < I$, so every X_j is a contraction. Hence $\|X^w\| \leq 1$ for every word w , and

$$\|\text{row}(X^w)_{|w|=N}\|^2 = \left\| \sum_{|w|=N} X^w X^{w*} \right\| \leq d^N.$$

Thus $A_{N,d} \leq d^{N/2}$.

For the reverse inequality, let $\mathcal{F}_{d,N}$ be the finite-dimensional Hilbert space with orthonormal basis $(e_v)_{|v| \leq N}$ indexed by words in $\{1, \dots, d\}$, including the empty word $\emptyset$. Define truncated left creation operators by

$$L_j e_v = \begin{cases} e_{jv}, & |v| < N, \\ 0, & |v| = N. \end{cases}$$

The ranges of the L_j are mutually orthogonal and sum to the non-vacuum subspace. Given $0 < r < 1$, set $X_j = r L_j^*$. Then

$$\sum_{j=1}^d X_j^* X_j = r^2 \sum_{j=1}^d L_j L_j^* = r^2 (I - P_\emptyset) < I,$$

so $X \in \mathfrak{C}_d$ at the finite matrix level $1 + d + \dots + d^N$.

For each word $w = (w_1, \dots, w_N)$, the vector $X^{w*} e_\emptyset$ is $r^N e_{w_N \dots w_1}$, and applying X^w sends it back to $r^{2N} e_\emptyset$. Therefore

$$\sum_{|w|=N} X^w X^{w*} e_\emptyset = d^N r^{2N} e_\emptyset,$$

and the word-row norm is at least $r^N d^{N/2}$. Letting $r \uparrow 1$ proves $A_{N,d} \geq d^{N/2}$. $\square$

4 Counterexample to right regularity

Theorem 2. *The nc column ball $\mathfrak{C}_d$ is right regular if and only if $d = 1$. In particular, it is not right regular for every $d \geq 2$.*

Proof. Fix $d \geq 2$ and, for every $N \geq 1$, consider

$$f_N(Z) = Z_1^N.$$

This monomial lies in the ideal generated by the words of length N . Since the first coordinate of every $X \in \mathfrak{C}_d$ is a contraction, $\|f_N\| \leq 1$. Conversely, scalar points $(r, 0, \dots, 0)$ with $r \uparrow 1$ show that $\|f_N\| = 1$.

The order- N right Taylor–Taylor expansion of f_N is simply

$$f_N(Z) = \sum_{|w|=N} Z^w \Delta^w f_N(0, \dots, 0, Z) = Z_1^N \cdot 1.$$

Thus the coefficient column has one entry equal to the constant function one and every other entry zero. Its supremum norm on $\mathfrak{C}_d$ is exactly one. By Lemma 1, the left side of (1) for f_N is therefore

$$d^{N/2} \cdot 1 = d^{N/2}.$$

No constant M independent of N can satisfy (1). Hence $\mathfrak{C}_d$ is not right regular.

When $d = 1$, the column ball is the one-variable row ball, which is right regular by the row-ball example immediately following Definition 5.5 in the source paper [1]. $\square$

5 Verification

The proof uses finite matrices only. A numerical checker included with this packet constructs the truncated shifts for $d = 2, 3$ and $N = 1, \dots, 4$ at $r = 0.83$. It verifies

$$\left\| \sum_j X_j^* X_j \right\| = r^2, \quad \|\text{row}(X^w)_{|w|=N}\| = r^N d^{N/2}.$$

All cases pass. These finite checks illustrate but do not replace the exact argument above.

6 Scope and novelty status

The theorem fully answers Example 5.9 and gives the exact word-row growth. It does not characterize right regularity for general nc operator balls.

A bounded search on 11 August 2026 checked the four run indexes, the current arXiv v2 and journal metadata, the exact phrases “column unit ball” and “right regular”, and combinations with “nc operator ball”, “truncated shift”, and the growth $d^{N/2}$. No separate answer was found. This was not an exhaustive citation search, so novelty remains unconfirmed.

7 Human-review recommendation

High priority. The key audit is that the regularity constant in Definition 5.5 is uniform in N . Under that stated quantifier, Lemma 1 and the monomial Z_1^N give an immediate obstruction.

References

- [1] Jeet Sampat and Orr Moshe Shalit, *Weak-* and completely isometric structure of noncommutative function algebras*, J. Math. Anal. Appl. **550** (2025), 129552, arXiv:2411.09372v2, doi:10.1016/j.jmaa.2025.129552.